

Snowflake Cloud Database

Quick Study Guide | architecture, SQL, governance, and exam-ready reminders

The 30-second model	What to remember
Cloud data platform	Storage, compute, and cloud services are independently scalable.
Virtual warehouse	A compute cluster that executes SQL; it does not store data.
Database / schema / table	Logical hierarchy: database > schema > object.

1. Core architecture

Layer	Purpose	Study cue
Database storage	Compressed, encrypted columnar data in cloud object storage.	Snowflake manages files and micro-partitions.
Compute	Virtual warehouses run queries, load, unload, and transform.	Size changes power; multi-cluster changes concurrency.
Cloud services	Authentication, metadata, query optimization, access control, transactions.	Serverless functions use this layer.

2. Objects and data flow

Object	Use
Stage	Named location for files. Internal stages live in Snowflake; external stages point to cloud storage.
File format	Describes CSV, JSON, Parquet, etc.; reuse it in COPY commands.
COPY INTO	Loads files into tables or unloads query/table output to a stage.
View	Saved query. Secure views hide underlying definition and restrict optimization details.
Stream + task	Stream tracks DML changes; task schedules SQL or creates a dependency chain.

3. Performance essentials

Feature	When / why
Auto-suspend	Stops idle warehouse to control cost. Auto-resume starts it on demand.
Result cache	Reuses a prior result when the query and data are unchanged; no warehouse required for cache retrieval.
Micro-partition pruning	Snowflake skips partitions using metadata. Filter on selective columns; avoid needless transforms in predicates.
Clustering key	Optional; useful for very large tables with frequent selective filters / joins. Not a default fix.
Scaling	Resize warehouse for faster single queries; multi-cluster warehouse for many concurrent queries.

SQL & Operations Cheat Sheet

Memorize the intent of each command, not just its syntax.

Create & use	<pre>CREATE DATABASE analytics; USE DATABASE analytics; CREATE SCHEMA raw;</pre>
Warehouse	<pre>CREATE WAREHOUSE etl_wh WAREHOUSE_SIZE = 'SMALL' AUTO_SUSPEND = 60 AUTO_RESUME = TRUE; USE WAREHOUSE etl_wh;</pre>
Load files	<pre>CREATE STAGE my_stage; COPY INTO customers FROM @my_stage/customers/ FILE_FORMAT = (TYPE = CSV SKIP_HEADER = 1);</pre>
Semi-structured	<pre>SELECT src:customer.name::STRING, src:total::NUMBER FROM events; -- VARIANT stores JSON-like data</pre>
Time Travel	<pre>SELECT * FROM orders AT (OFFSET => -60*10); -- Query/restore recent historical data</pre>

4. Security and governance

Concept	Key point
RBAC	Privileges are granted to roles; roles are assigned to users or other roles. Prefer least privilege.
System roles	ACCOUNTADMIN is powerful; separate day-to-day roles such as SYSADMIN, SECURITYADMIN, USERADMIN.
Managed access schema	Object owners cannot grant privileges; the schema owner / managed-access role centralizes grants.
Masking policy	Dynamically masks sensitive column values based on role/context.
Row access policy	Filters which rows a role can see; often maps users/roles to permitted values.
Tags	Classification and metadata labels; can support governance and policy assignment.

5. Data protection & sharing

Feature	Remember
Time Travel	Historical data access for accidental changes. Retention depends on edition/object settings.
Fail-safe	Snowflake-managed disaster recovery period after Time Travel; not user-queryable.
Zero-copy cloning	Fast logical clone that initially shares underlying storage; changes create new storage only as needed.
Secure Data Sharing	Shares live data without copying it. Consumers can query a shared database.
Replication	Replicates databases/account objects across regions or cloud platforms for continuity / locality.

Fast Review

Use these prompts for a 5-minute final pass.

Fast self-check

Question	Answer
Need more concurrent users?	Use multi-cluster warehouse (or workload separation).
Need faster heavy query?	Increase warehouse size, then inspect query profile / pruning.
Need to undo a DELETE?	Use Time Travel if within retention; Fail-safe is not self-service.
Need share live data externally?	Secure Data Sharing, not export-and-send files.
Need incremental pipeline?	Stream for changes + task to process on a schedule/dependency.

Memory hooks

Cue	Recall
Warehouse = engine	It provides compute only. Data remains in Snowflake storage.
Resize vs. multi-cluster	Resize helps a query; multi-cluster helps a queue of users / queries.
Stream = what changed	A stream exposes data-change records, usually consumed by a task.
Clone = copy without copying	Zero-copy cloning is fast because it initially shares storage.
Share = live, not exported	Secure Data Sharing gives consumers access to current provider data.

Study tip: draw the three layers from memory, then explain how a warehouse, stage, stream, and task fit into one ingestion pipeline.